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SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Influence of physical, chemical, thermal, biological and temporal process to Time-to-Failure

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Concept of P-F Curve

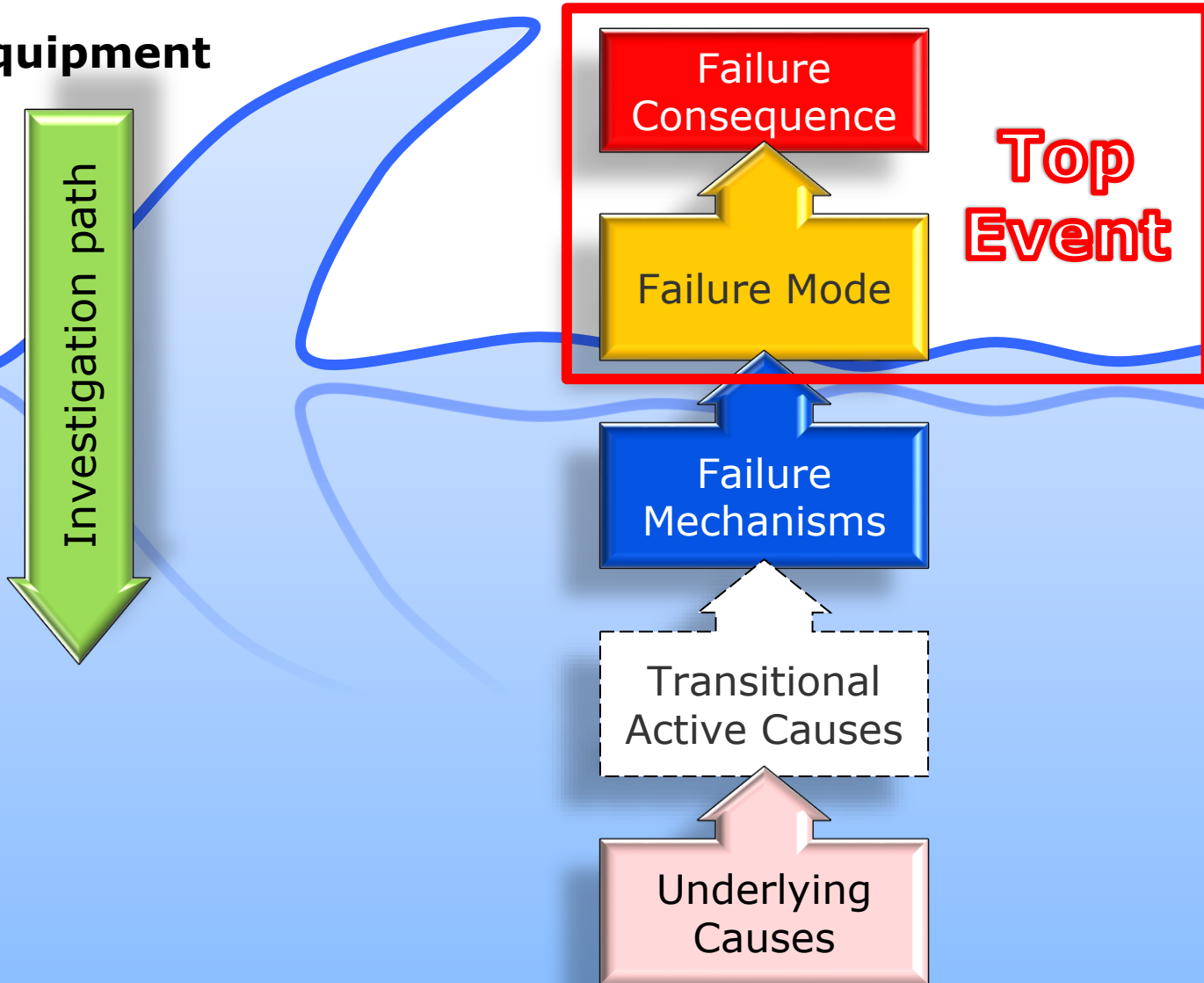
Learning Objectives

By the end of this session, you should be able to:

- Understand the concept of P-F Curve.
- Understand the influence of physical, chemical, thermal, biological and temporal process to Time-to-Failure

Introduction to Failure Causation Model

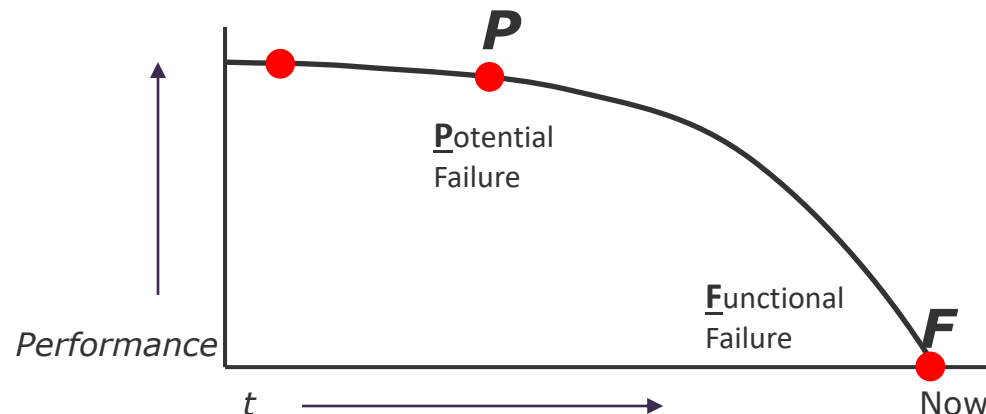
How equipment fails...



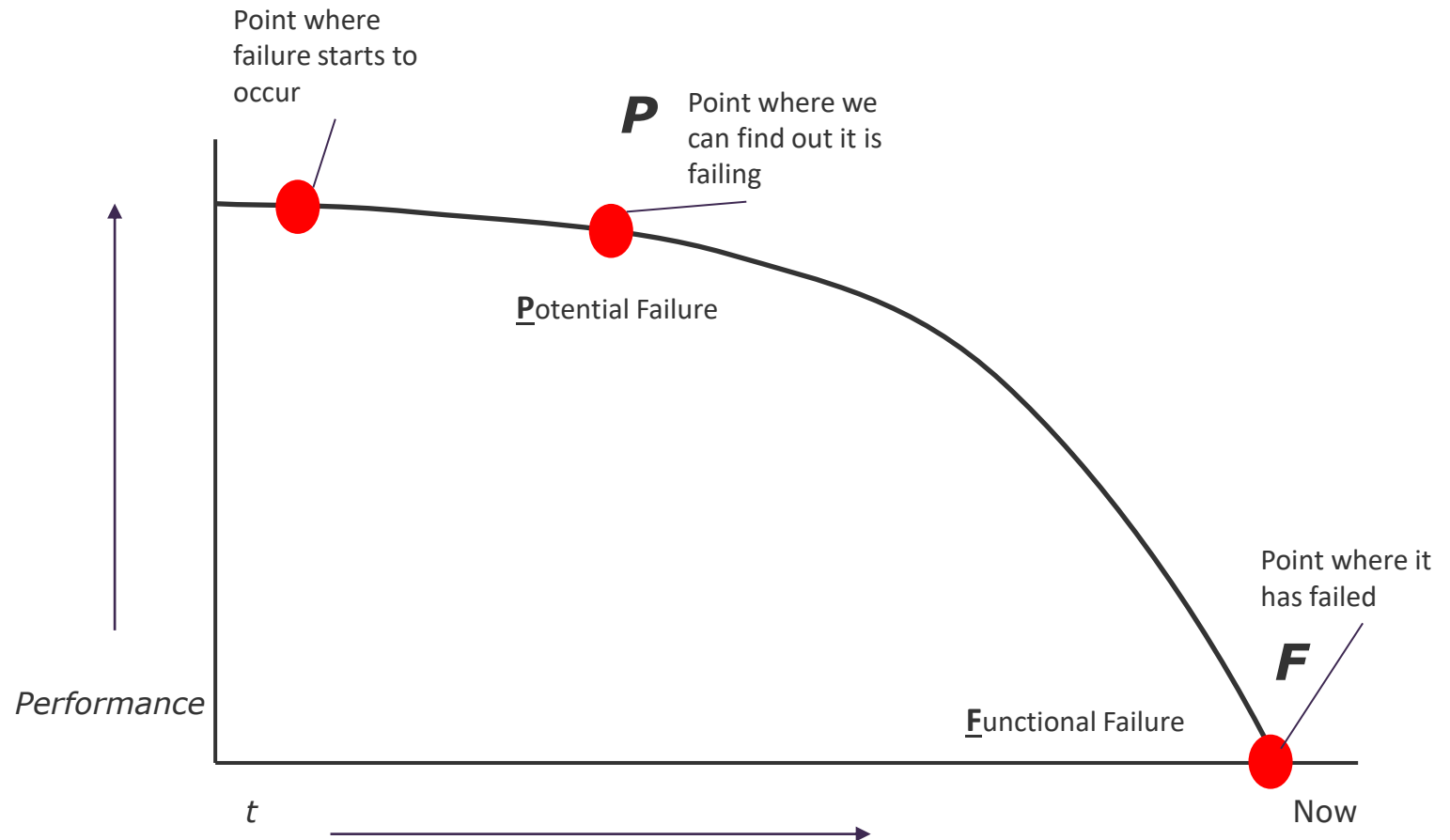
Introduction to P-F Curve

What is P-F curve?

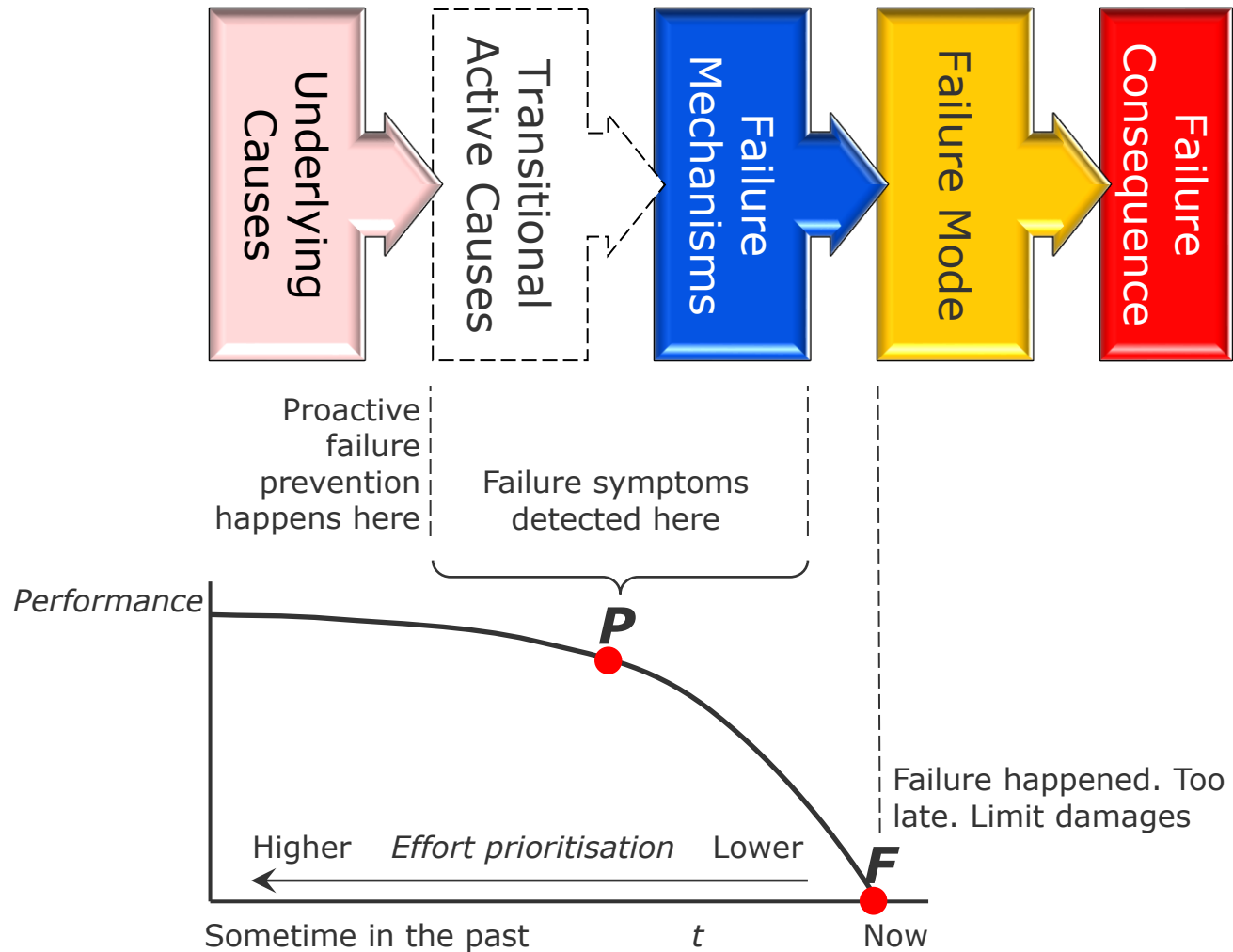
- The P-F curve is a common way to represent the behaviour of an asset, or asset component before actual functional failure has occurred.
- Failure is a **PROCESS**, not an EVENT.
- Common failure mechanisms leading to failure like wear, stretch, misalignment do not lead to failure immediately.
- The concept is important to understand the need for Predictive Maintenance.



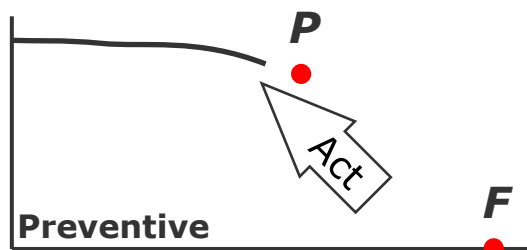
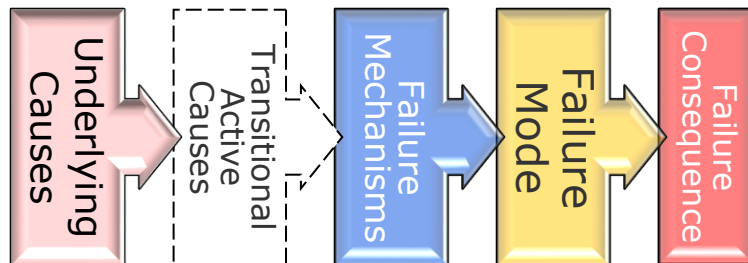
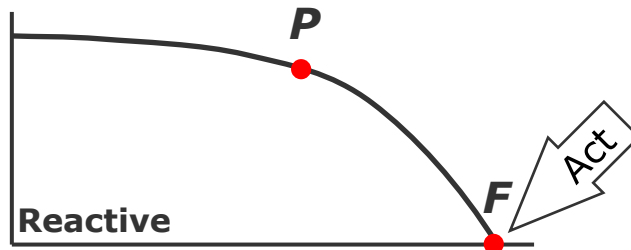
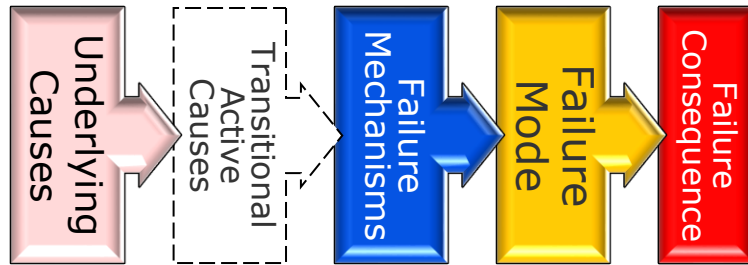
Let's get to know P-F Curve



The P-F (Potential to Failure) Curve and Failure Causation Model (1 of 4)

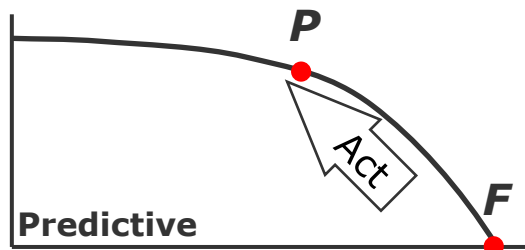
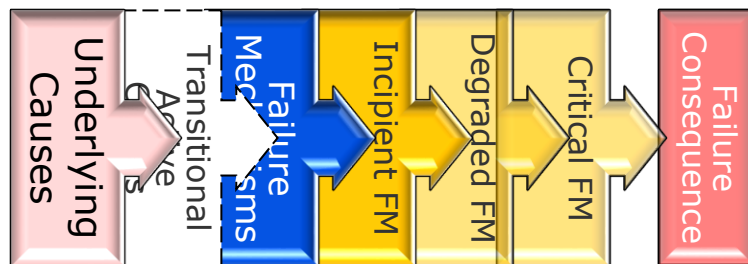
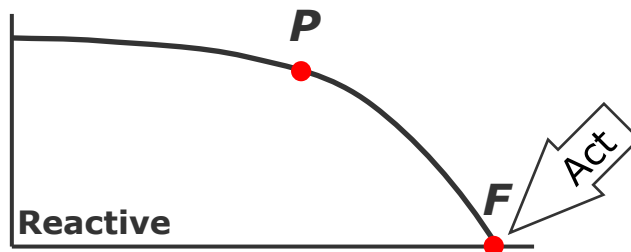
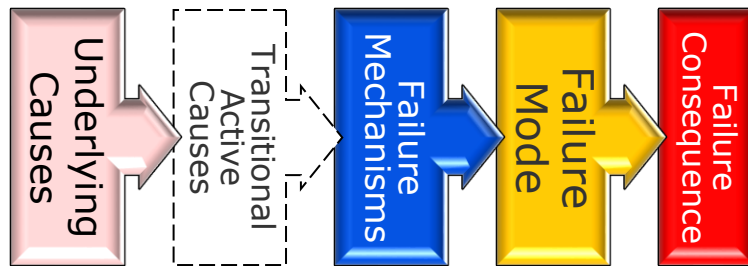


The P-F (Potential to Failure) Curve and Failure Causation Model (2 of 4)



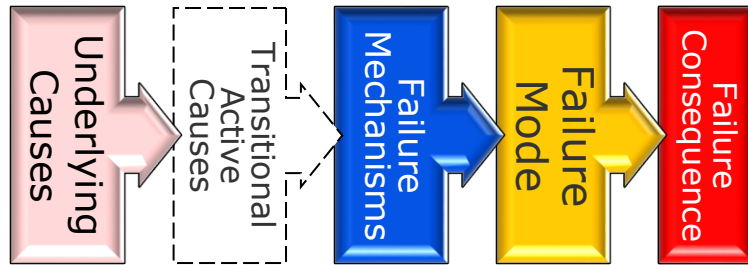
- Organisation that has more serious systemic issues.
 - Failures will happen earlier.
 - Consequences are often unpleasant.
- The same organisation adopting preventive tasks during operation phase, removes/corrects the failing item at fixed intervals.
- Failure Consequence is avoided, at the expense of higher maintenance cost, manpower fatigue, and *risk* of induced failures.

The P-F (Potential to Failure) Curve and Failure Causation Model (3 of 4)

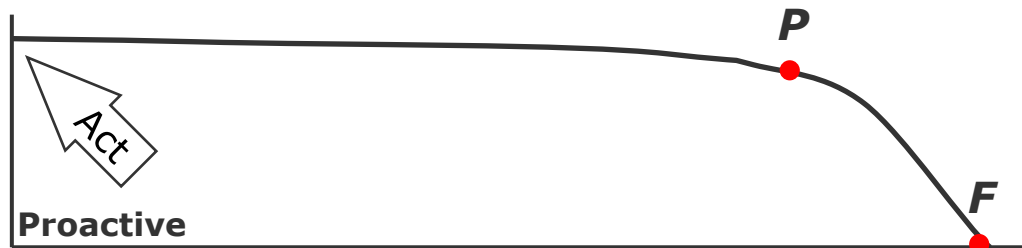
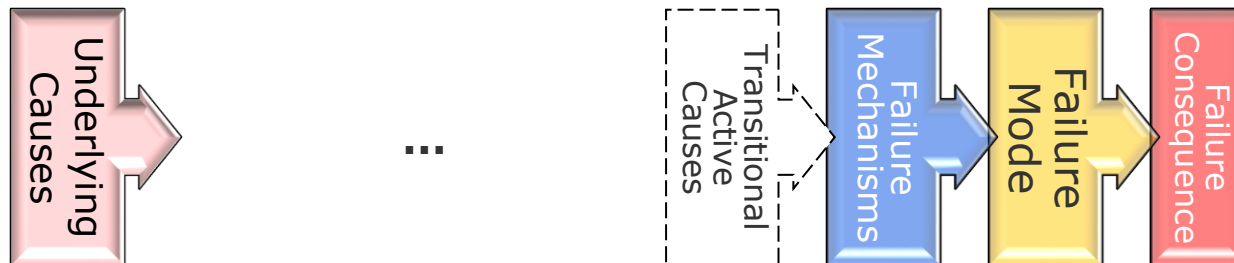
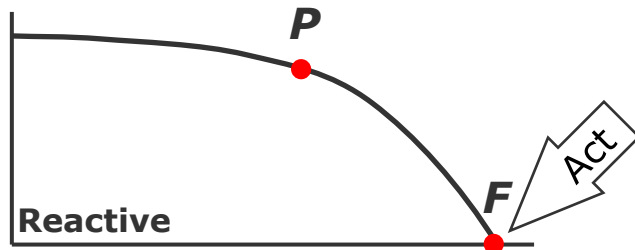


- The same organisation adopting predictive tasks, identifies the Failure Mode at incipient level, and address it just-in-time prior to failure.
- Failure Consequence is avoided, but the Underlying Causes still hang on.
- Tools, skills and discipline must be maintained good.
- A missed inspection would result in a critical failure.

The P-F (Potential to Failure) Curve and Failure Causation Model (4 of 4)



- A better organisation knows how failures (and poor performances) happen, proactively prepares for those, experiences failures later, with low consequences.



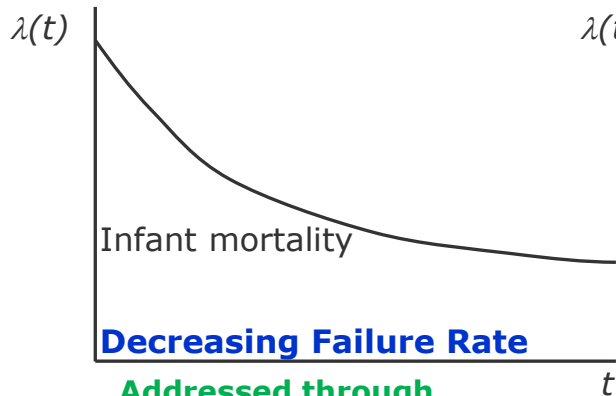
The P-F Curve and The Bathtub Curve

(Usually) caused by...

- Design,
- fabrication, and
- installation errors (*quality assurance related*).

- Random loads from operation errors, and
- maintenance errors, and
- environment and chance events.

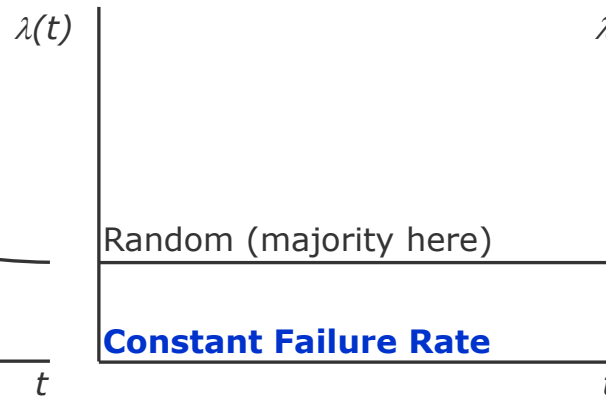
- Normal aging, or
- abnormal aging.



Addressed through...

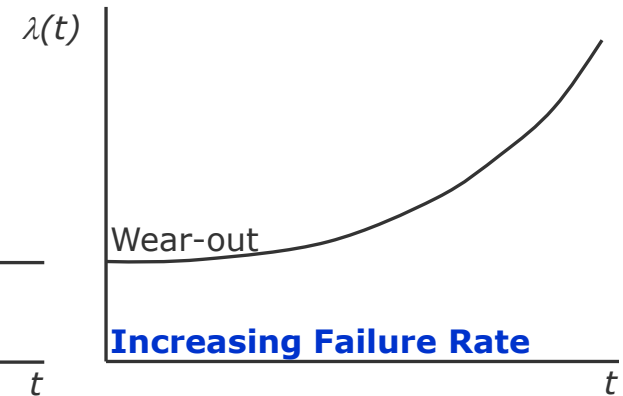
- QA & QC,
- acceptance testing,
- burn-in operation,
- (may combine with) inspection and *testing* (for *hidden failures*), and/or
- Benchmarking and/or RCFA.

NOT by more Preventive Maintenance.



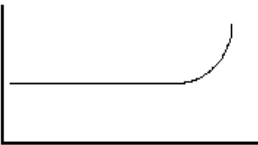
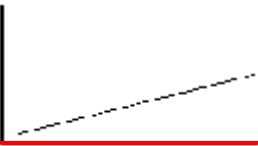
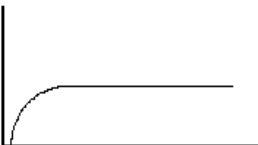
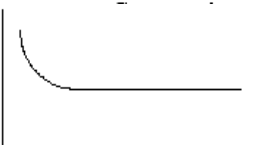
- Inspection, for observable P-F,
- Testing and/or redundancy, for hidden failures,
- redundancy, for evident failures with no observable P-F,
- better integrity,
- following procedures, and/or
- Benchmarking and/or RCFA.

NOT by more Preventive Maintenance.



- Preventive Maintenance,
- rerating, and/or
- Benchmarking and/or RCFA.

About 71-92% of failures are random

	1968 UAL*	1973 Broberg	1982 MSP**	2001 SUB- MEPP** *	
	4%	3%	3%	2%	Bathtub curve.
	2%	1%	17%	10%	Traditional wear-out pattern expected.
	5%	4%	3%	17%	Constant increase in failure rate.
	7%	11%	6%	9%	Low rate of failures when new, then rapid increase to a constant level, regardless of age (random failure).
	14%	15%	42%	56%	Constant failure rate, regardless of age (random failure).
	68%	66%	29%	6%	High "infant mortality" initially, followed by random failure .

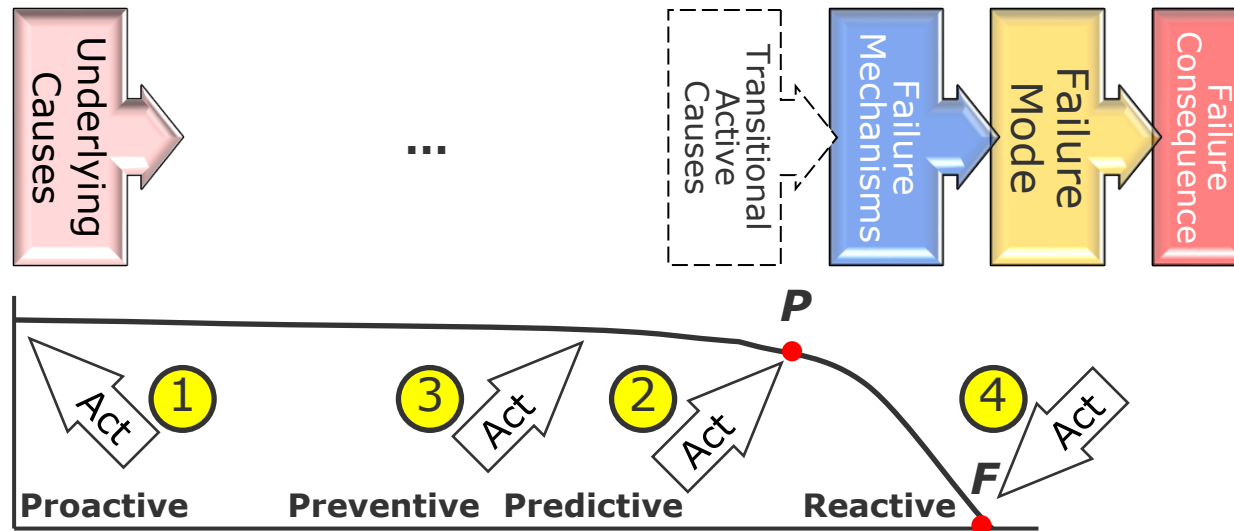
Source: Moubray, J. (2007)

*United Airlines

**US Navy

***Submarine Maintenance Engineering Planning and Procurement

Summary: Understanding and applying The P-F Curve, and the failure patterns



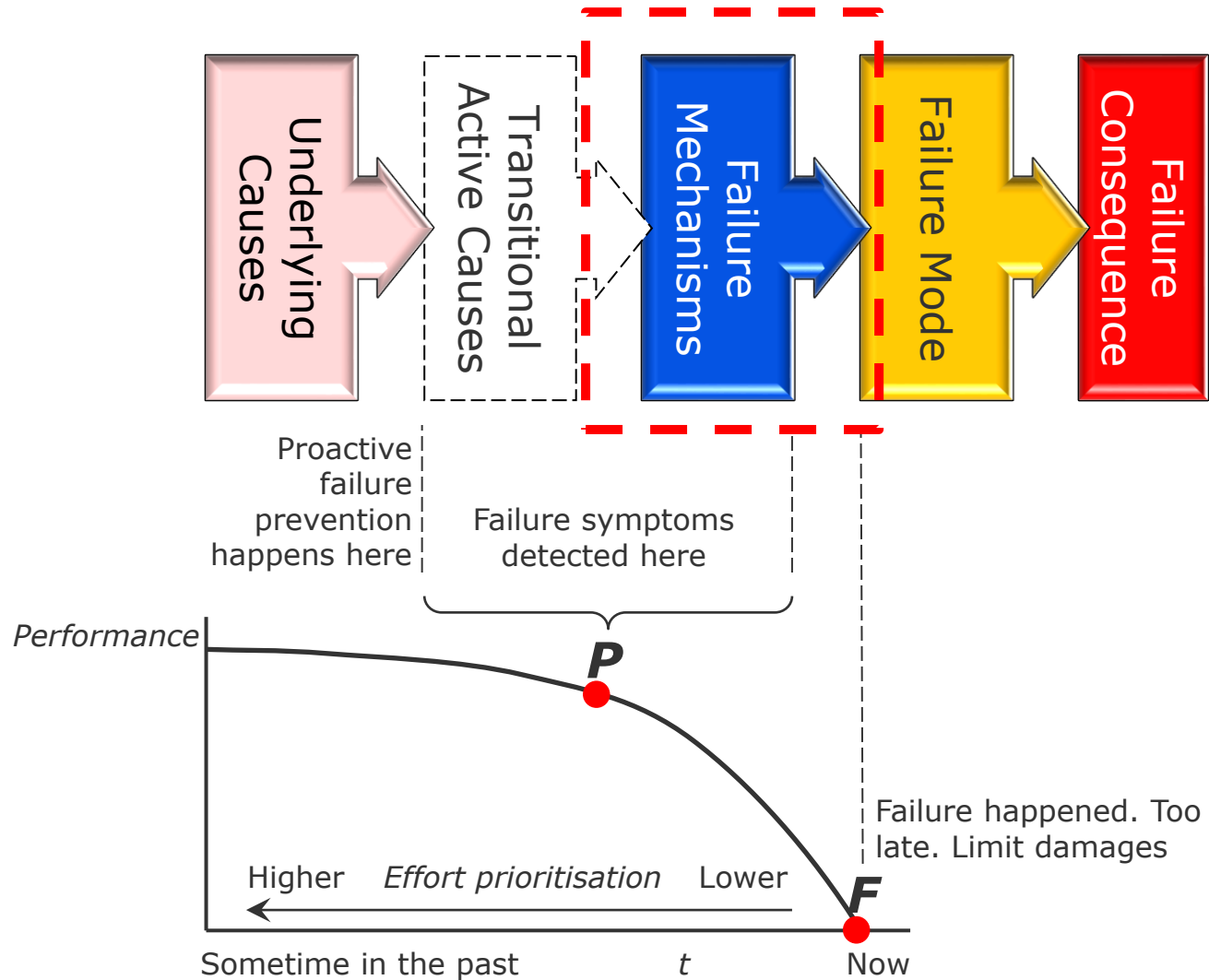
1. Focus on proactive tasks:
 - a. **Design it right** – meeting the intended functions,
 - b. **Operate it right** – following the procedures and operating limits,
 - c. **Maintain it right** – following the procedures.
2. ... followed by predictive tasks:
 - a. assess system performance
3. ... be choosy on Preventive Maintenance tasks, and
4. ... keep reactive maintenance as low as possible.



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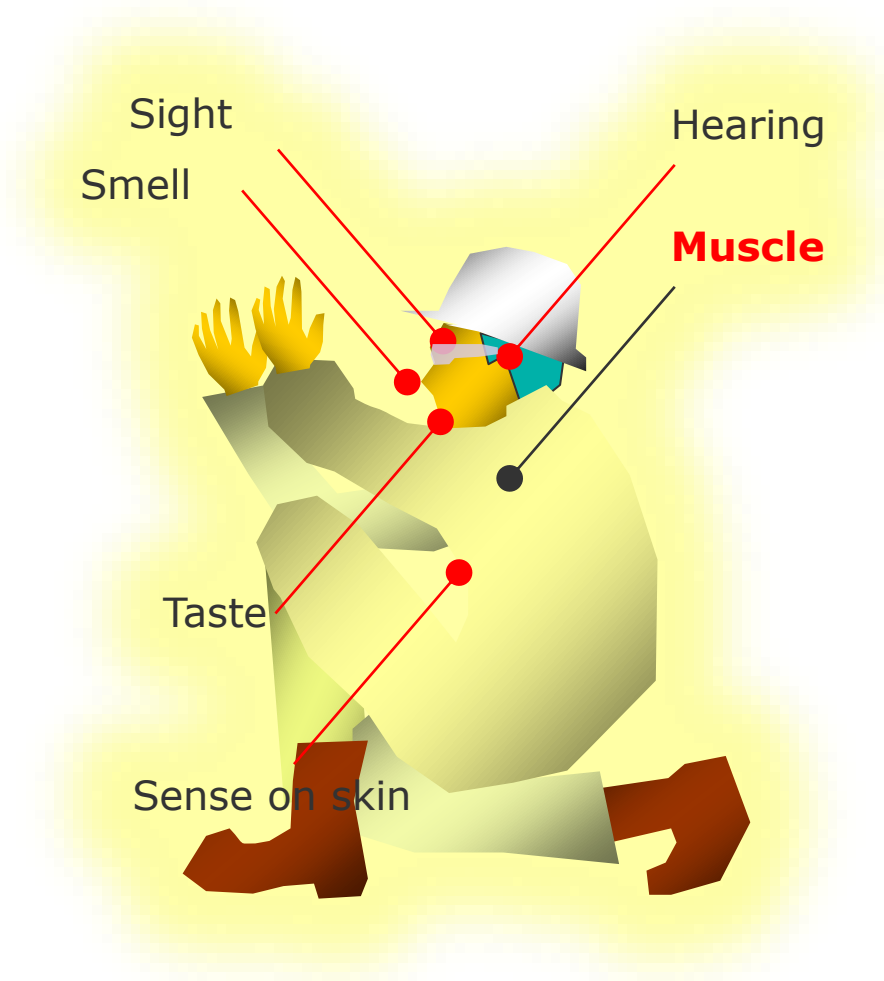
Influence of physical, chemical, thermal, biological and temporal process to Time-to-Failure

The P-F (Potential to Failure) Curve and Failure Causation Model



Human senses as a tool to “check” failure mechanisms

- “Extension” of our senses and limbs.



How do we use our senses to identify Potential Failure (1 of 13)

Introducing the word.....

PROGNOSTIC

The process of predicting the future reliability of an asset by assessing the extent of deviation or degradation of the asset from its expected normal conditions¹

Note 1: Michael G Pecht , “Health Management of Electronics”

How do we use our senses to identify Potential Failure (2 of 13)

Further discussion on PROGNOSTICS.....

The prognostics of an asset can give advance warning of impending failure and assist us in taking the appropriate measures

This will prevent any catastrophic failures and reduce unscheduled maintenance.

This has also become the preferred approach to achieve efficient maintenance and reduce the life cycle costs.

How do we use our senses to identify Potential Failure (3 of 13)

If we can access and know the extent of deviation and degradation of an asset, the information can be used for the following objectives:

- ✓ Provide advanced warning of system failures
- ✓ Enable condition-based maintenance
- ✓ Increase system availability through an extension of maintenance cycles and/ or timely repair
- ✓ Lower life-cycle cost of equipment from reductions in inspection costs, downtime & inventory

How do we use our senses to identify Potential Failure (4 of 13)

Prognostics

Time-to-Failure

Failure mechanisms

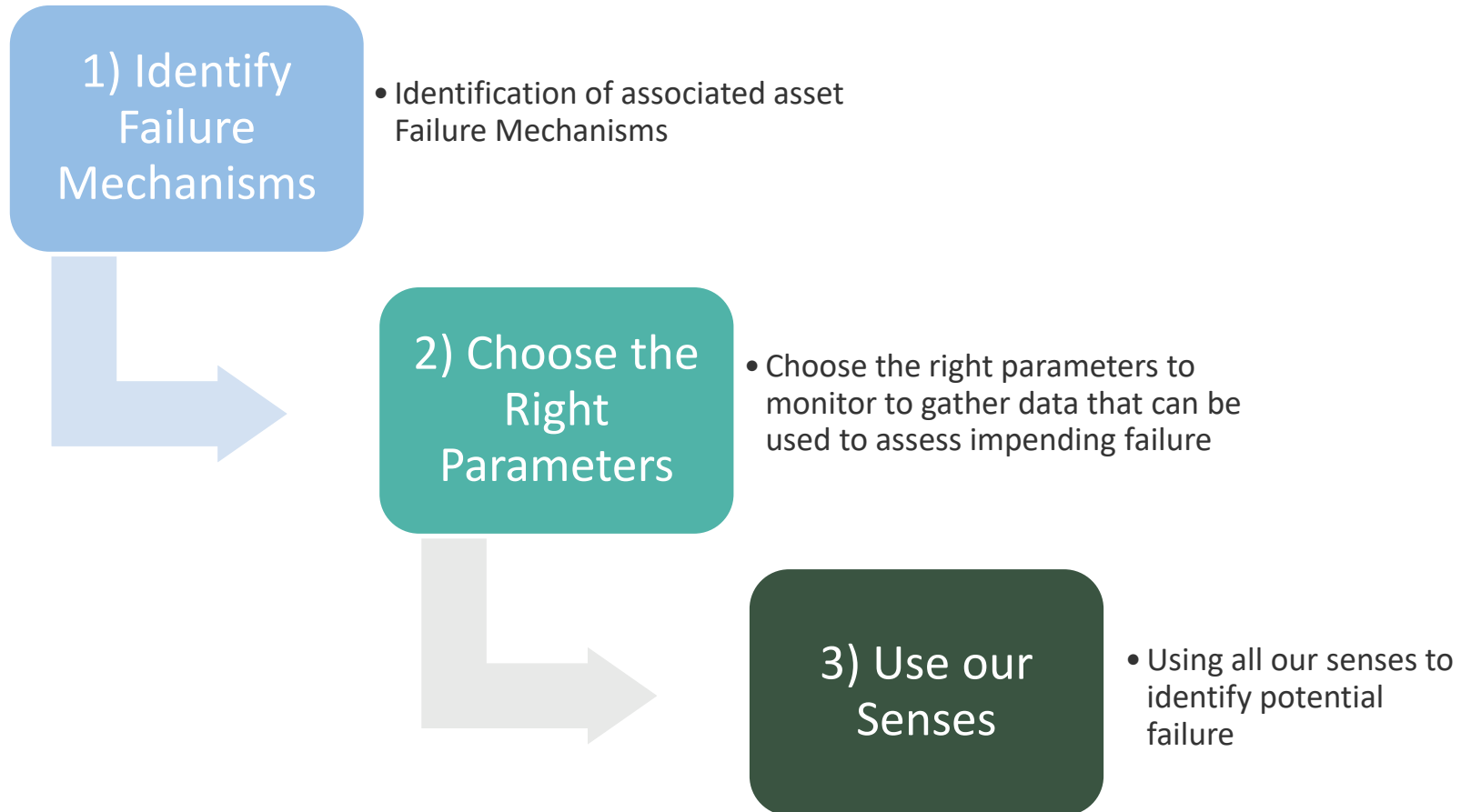
How are these terms related to each other?

How do we use our senses to identify Potential Failure (5 of 13)

The key requirement for *time-to-failure* is knowing what are the *failure mechanisms* associated to the asset, hence we could “predict” (remember *PROGNASTICS*?) when will the failure happen (or might not happen!)

Having this understanding pertaining time-to-failure of an asset, what will be our next step?

How do we use our senses to identify Potential Failure (6 of 13)



How do we use our senses to identify Potential Failure (7 of 13)

1) Identify Failure Mechanisms

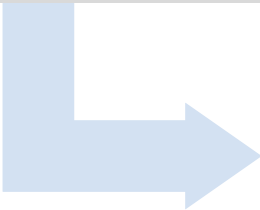
- Identification of associated asset Failure Mechanisms

How do we use our senses to identify Potential Failure (8 of 13)

Failure Mechanism	Description of the failure mechanism
Leakage	External and internal leakage
Vibration	Abnormal vibration
Alignment failure	Failure caused by faulty clearance or alignment
Corrosion	All types of corrosion, both wet (electrochemical) and dry (chemical)
Erosion	Erosive wear
Wear	Abrasive and adhesive wear e.g. scoring, galling, fretting
Breakage	Fracture, breach, crack
Fatigue	Could be a cause of breakage
Overheating	Material damage due to overheating/ burning

For a full listing of Failure Mechanism notifications, refer to ISO 14224 (pages 115-116)

How do we use our senses to identify Potential Failure (9 of 13)



2) Choose the Right Parameters

- Choose the right parameters to monitor to gather data that can be used to assess impending failure

How do we use our senses to identify Potential Failure (10 of 13)

Failure Mechanism	Description of the failure mechanism	Parameters
Leakage	External and internal leakage	wall thickness
Vibration	Abnormal vibration	temperature, sound, load
Alignment failure	Failure caused by faulty clearance or alignment	temperature, sound, load
Corrosion	All types of corrosion, both wet (electrochemical) and dry (chemical)	wall thickness
Wear	Abrasive and adhesive wear e.g. scoring, galling, fretting	age
Breakage	Fracture, breach, crack	pressure, temperature
Fatigue	Could be a cause of breakage	age
Overheating	Material damage due to overheating/ burning	temperature

How do we use our senses to identify Potential Failure (11 of 13)



3) Use our Senses

- Using all our senses to identify potential failure

How do we use our senses to identify Potential Failure (12 of 13)

Failure Mechanism	Description of the failure mechanism	Parameters	Senses
Leakage	External and internal leakage	wall thickness	sight
Vibration	Abnormal vibration	temperature, sound, load	sight, hearing, smell
Alignment failure	Failure caused by faulty clearance or alignment	temperature, sound, load	sight, hearing, smell
Corrosion	All types of corrosion, both wet (electrochemical) and dry (chemical)	wall thickness	sight, sense of skin
Wear	Abrasive and adhesive wear e.g. scoring, galling, fretting	age	sight, hearing
Breakage	Fracture, breach, crack	pressure, temperature	sight, hearing
Fatigue	Could be a cause of breakage	age	sight
Overheating	Material damage due to overheating/ burning	temperature	sight, sense of skin

How do we use our senses to identify Potential Failure (13 of 13)

1) Identify Failure Mechanisms

- Identification of associated Failure Mechanisms

Predictive Maintenance

Parameters to
data that can be
pending failure

3) Use our Senses

- Using all our senses to identify potential failure



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Thank you